

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-13 06:30 UTC

SEC P vs NP — notebook summary

Generated 2026-05-13 06:30 UTC

- Cycles recorded: **699**
- Time span: 464.8h (~1.50 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	662
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	39
Schur-Weyl Duality	26
Free Probability Theory	18
Representation Theory of Symmetric Groups	17
Matroid Theory	16
Free Probability	14
Additive Combinatorics	13
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Persistent Homology	5
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4

Field	Cycles
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-13 00:43 UTC	INCONCLUSIVE	Plethysm Coefficient Gap in Symmetric Powers of Permanent vs Dete
2026-05-13 01:20 UTC	INCONCLUSIVE	Negative Eigenvalue Count in SOS Moment Matrix Bounded by Log-Siz
2026-05-13 01:46 UTC	INCONCLUSIVE	Free Cumulant Norm Gap in Read-Twice BPs for IP_2
2026-05-13 02:16 UTC	INCONCLUSIVE	Secant Variety Dimension Bounds Disjointness Communication Rank
2026-05-13 02:43 UTC	INCONCLUSIVE	Free Cumulant Spread Bounds Disjointness Communication Complexity
2026-05-13 03:13 UTC	INCONCLUSIVE	Schur-Weyl Decomposition Term Count vs. Monotone Circuit Size for
2026-05-13 03:38 UTC	INCONCLUSIVE	Free Cumulant Sum Inversely Proportional to Disjointness Communic
2026-05-13 04:06 UTC	INCONCLUSIVE	Free Cumulant Spread Bounds Disjointness Communication Complexity
2026-05-13 04:27 UTC	INCONCLUSIVE	Sum-Product Complexity Lower Bounds for ACC ⁰ Circuits
2026-05-13 04:40 UTC	INCONCLUSIVE	Free Cumulant Sum Lower Bounds Randomized Communication Complexit
2026-05-13 04:58 UTC	INCONCLUSIVE	Toric Variety Degree Inverse to SOS Refutation Degree for 3-SAT

Time	Verdict	Title
2026-05-13 05:57 UTC	INCONCLUSIVE	Persistent Homology Total Persistence Inversely Proportional to R
2026-05-13 06:08 UTC	INCONCLUSIVE	Plethysm Coefficient Sum Gap in Symmetric Powers of Permanent vs
2026-05-13 06:23 UTC	INCONCLUSIVE	Gröbner Basis Step Count vs. Extended Frege Proof Length for 3-SA
2026-05-13 06:30 UTC	INCONCLUSIVE	Free Cumulant Sum Gap in Read-Twice BPs for 3-SAT

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```